

Ali Mohsen

✉ ali2moh.04@gmail.com | 📞 +961 76 683 027 | 🌐 github.com/amoh909 | 🔗 linkedin.com/in/alimohsen5

PROFESSIONAL PROFILE

Computer Science graduate from the American University of Beirut with a Mathematics minor and strong foundations in full-stack software engineering, machine learning, and computational systems. Skilled in building end-to-end applications using React, TypeScript, Django REST Framework, PostgreSQL, and Docker, with hands-on experience in ML systems, GPU computing, and algorithm optimization.

TECHNICAL SKILLS

Languages: Python, Java, C/C++, JavaScript/TypeScript, SQL

Fullstack Development: Spring Boot, React, Node.js, Django, Tailwind CSS, MUI, shadcn/ui, TanStack

Databases & APIs: PostgreSQL, SQLite, REST APIs, External API Integration, Data Ingestion Pipelines

Machine Learning & AI: PyTorch, TensorFlow, Scikit-learn, NumPy, n8n — CNNs, U-Net, Anomaly Detection

Systems & Cloud: CUDA, Parallel Computing, GPU Optimization, Docker, Git, AWS, Linux

WORK EXPERIENCE

Quantum Trading Company

Aug 2026 – Present

Software Engineer Intern

- Developed and customized core business modules and workflows using Python and enterprise management frameworks.
- Implemented custom backend logic, database integrations, and tailored user interfaces to optimize business operations.
- Participated in system testing, code reviews, and maintenance to ensure high performance and application stability.

Ciatek IT & Software Solutions, Lebanon

Jun 2025 – Aug 2025

Frontend Developer Intern

- Developed and maintained responsive web interfaces using React, TypeScript, Tailwind CSS, and shadcn.
- Built reusable components and implemented frontend features following modern component-based architecture.
- Collaborated with engineering teams to integrate UI components and improve overall application usability.

EDUCATION

American University of Beirut

Aug 2023 – May 2026

B.S. in Computer Science, Minor in Mathematics

GPA: 3.59/4

Relevant Coursework: Object Oriented Programming, Data Structures and Algorithms, Computer Organization & Design, Software Engineering, Database Systems, Operating Systems, Machine Learning, GPU Computing, Optimization

PROJECTS & RESEARCH

- **MENA EconLens: Economic Intelligence Dashboard** 2026
 - Built a full-stack economic analytics platform using Django REST Framework, React, TypeScript, Tailwind CSS, PostgreSQL, Docker, and Recharts.
 - Implemented World Bank data ingestion, normalized country-indicator-year models, and linear-regression forecasting.
- **Healthcare Provider Fraud Detection System** 2026
 - Built an end-to-end anomaly detection pipeline analyzing healthcare claims data to identify potentially fraudulent providers.
 - Conducted feature engineering and imbalance handling, training XGBoost/LightGBM classifiers for explainable risk scoring.
- **Brain Tumor Segmentation using U-Net** 2026
 - Designed and trained U-Net architecture for medical image segmentation on LGG MRI dataset in PyTorch.
 - Achieved ~0.82 Dice score and ~0.78 IoU on test set through custom loss function tuning and data augmentation.
- **Sparse Lower Triangular Matrix Solver (CUDA)** 2026
 - Implemented a CUDA-based parallel sparse triangular solver (SpTRSV) using GPU kernels and memory optimizations.
 - Achieved up to ~500× speedup over CPU baseline through parallel execution, level-set scheduling, and kernel optimization.

CERTIFICATES & LEADERSHIP

- **AWS Certified Cloud Practitioner (2025) | Neural Networks** — DeepLearning.AI
- **MOS Lebanon 2026** — 1st Place Excel
- **GDG AUB** — Software Development Team Member | **IEEE AUB Student Branch** — CMPS Representative
- **AUB** — CMPS 208 Lab Assistant
- **AUB Outlook** — Science & Tech Writer